

OLIST MARKETPLACE PERFORMANCE DASHBOARD

Business Intelligence Project Documentation

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Prepared for	BI Analyst / Dashboard Developer
Project period	September 2026
Platform	Microsoft Power BI
Dataset	Brazilian E-Commerce Public Dataset by Olist

Purpose: Provide a single analytical view of revenue, delivery, customer satisfaction, geography, product performance and seller performance using the Olist marketplace dataset.

1. Executive Summary

The Olist Marketplace Performance Dashboard was developed to provide a consolidated view of marketplace performance across sales, products, delivery, customer satisfaction, geography and sellers. The project brief identifies the need to replace manual spreadsheet-based reporting with a single source of truth that can be used by operations, sales and logistics stakeholders.

Overall dashboard snapshot

Metric	Observed value	Source page
Total Sales	13.59M	Executive Summary
Total Orders	99K	Executive Summary
Average Order Value	137.75	Executive Summary
Average Review Score	4.09	Executive Summary / Customer Satisfaction
On-Time Delivery	94.16%	Executive Summary / Delivery & Logistics
Delivered Orders	96K	Delivery & Logistics
Late Orders	8K	Delivery & Logistics
Average Delivery Days	12.41	Delivery & Logistics
Total Reviews	98K	Customer Satisfaction

Three key business insights

1. Revenue concentration

The dashboard shows that several categories account for a substantial share of sales. The leading displayed categories are Health & Beauty (~1.26M), Watches & Gifts (~1.21M), Bed, Bath & Table (~1.04M), Sports & Leisure (~0.99M), and Computers & Accessories (~0.91M). This indicates that category-level performance is an important lens for revenue management.

2. Geographic variation in delivery time

Overall on-time delivery is 94.16%, with approximately 96K delivered orders and 8K late orders. The state-level delivery visual shows substantial variation in average delivery time, ranging from about 9 days in SP to about 29 days in RR in the displayed dashboard. This indicates that logistics performance differs materially by geography.

3. Delivery performance and customer satisfaction should be analyzed together

The Customer Satisfaction and Delivery & Logistics pages show variation in delivery delay and average review score across states. The scatter plot provides a useful diagnostic view of this relationship. The dashboard supports further investigation of the association, but the visual alone should not be interpreted as proof that delivery delays directly cause lower ratings.

Important interpretation note: The 2017 YoY percentage shown in the Sales Performance Matrix is extremely high because 2016 contains only a partial period in the dataset. It should not be presented as a normal full-year growth comparison.

2. Business Context and Problem Statement

Olist connects small Brazilian retailers to major e-commerce marketplaces and handles logistics, payments and order fulfillment. The project brief states that approximately 100,000 orders from 2016–2018 were available, but reporting was being performed manually from spreadsheets.

Business problem

The project brief identifies limited visibility into which product categories and sellers drive revenue, where delivery performance breaks down, and how delivery performance relates to customer satisfaction.

Business objectives

- **Revenue & Growth:** analyze sales trends over time, category, region and seller.
- **Delivery Performance:** identify missed delivery estimates and delivery-time differences.
- **Customer Satisfaction:** analyze review scores in relation to delivery and category.
- **Payment Behavior:** understand payment methods and installment patterns.
- **Seller Performance:** analyze seller revenue, orders, delivery and reviews.
- **Geography:** compare Brazilian states/regions across sales and delivery metrics.

3. Data Sources

The project uses the Brazilian E-Commerce Public Dataset by Olist. The project brief identifies Kaggle as the source and lists the following CSV files.

Dataset	Purpose / grain
olist_orders_dataset.csv	Order-level lifecycle and date/status information
olist_order_items_dataset.csv	One item within an order; item price and freight
olist_products_dataset.csv	Product attributes and Portuguese category name
olist_customers_dataset.csv	Customer identity and customer location
olist_sellers_dataset.csv	Seller identity and seller location
olist_order_payments_dataset.csv	Payment transactions and installments
olist_order_reviews_dataset.csv	Customer reviews and review scores
olist_geolocation_dataset.csv	ZIP-level latitude/longitude and location information
product_category_name_translation.csv	Portuguese-to-English product category mapping

Dataset source identified in the project brief: Kaggle – Brazilian E-Commerce Public Dataset by Olist. The brief specifies CC BY-NC-SA 4.0 and states that the dataset is intended for personal/portfolio use rather than commercial resale.

4. Data Cleaning and Transformation

- **Duplicate handling:** complete duplicate rows were considered for removal only when every column was identical. Repeated business keys were retained when they were valid for the table grain.
- **Order items:** repeated order_id values were retained because an order can contain multiple items. The conceptual grain is order_id + order_item_id.
- **Payments:** repeated order_id values were retained because an order can contain multiple payment records.
- **Reviews:** blank review comments were treated as valid missing text rather than automatically invalid records.

- **Products:** product records were retained even where descriptive attributes were blank; missing values were not replaced with invented values.
- **Category translation:** the product category translation file was used as a mapping from Portuguese category names to English names.
- **Date preparation:** date fields were converted to appropriate date/date-time types and a dedicated Date table was created for time analysis.
- **Sales fact:** Orders and Order Items were combined into the sales fact structure used for revenue, quantity, freight and order-level analysis.
- **Separate facts:** Payments and Reviews were kept separate to avoid multiplying rows when combining datasets with different grains.

5. Data Model

The report uses a dimensional/relationship-based model in which sales is maintained at order-item grain while payment and review data remain separate. An order-level table is used to connect order-related facts, while product, seller, customer and date dimensions provide analytical filtering.

Dimension / bridge	Relationship target	Purpose
Date	facts_sales	Time filtering and YTD/MTD/QTD analysis
Products	facts_sales	Product and category analysis
Sellers	facts_sales	Seller revenue/order analysis
Customers	olist_orders_dataset	Customer state and location filtering
olist_orders_dataset	facts_sales	Order-level bridge to sales items
olist_orders_dataset	facts_payments	Order-level payment connection
olist_orders_dataset	facts_reviews_dataset	Order-level review connection

Modeling principle: repeated keys are not automatically duplicates. Relationships were designed around table grain to reduce accidental row multiplication, especially for payments and reviews.

6. Key DAX Measures

Measure	Logic / purpose
Total Sales	SUM(facts_sales[price])
Total Orders	DISTINCTCOUNT(facts_sales[order_id])
Total Quantity	COUNTROWS(facts_sales)
Average Order Value	DIVIDE([Total Sales], [Total Orders])
Sales YTD / MTD / QTD	Time-intelligence measures using the Date table
Sales Previous Year	CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
Sales YoY Growth %	Current sales versus previous-year sales
Average Actual Delivery Days	Order-level average using purchase date and delivered date
Average Estimated Delivery Days	Order-level average using purchase date and estimated delivery date
On-Time Delivery %	On-time delivered orders divided by delivered orders
Late Orders	Delivered orders whose delivery date is later than the estimated date
Average Review Score	AVERAGE(facts_reviews_dataset[review_score])
Total Reviews	DISTINCTCOUNT(facts_reviews_dataset[review_id])

Measure	Logic / purpose
Total Payment Value	SUM(facts_payments[payment_value])
Average Payment per Order	Payment value divided by distinct paid orders

Important AOV design choice: AOV uses total sales divided by distinct orders rather than AVERAGE(price), because the sales fact is at order-item grain and a single order can contain multiple items.

7. Dashboard Structure

Page	Content
Executive Summary	Revenue, orders, AOV, review score, on-time delivery, monthly revenue trend, order status, category performance and sales by year.
Sales & Product Performance	Monthly revenue trend, price vs freight cost, top/bottom product performance and sales performance matrix.
Delivery & Logistics	Actual vs estimated delivery time, delivery delay vs review score, delivery days by state, on-time/late delivery and cancelled orders.
Customer Satisfaction	Review score distribution, low-rating reviews by state, delivery delay vs review score and average review score by category.
Geography Analysis	Sales by state, orders by state, state-level delivery time and geographic map.
Seller Performance	Top sellers by sales and orders plus seller revenue/review-score analysis.

Interactivity implemented: State, Product Category and Year slicers are visible across the report pages, and a Reset/Clear Slicer control is included for returning filters to the default state.

Scope note: The project brief lists a dedicated Payments page as a required page. The final six-page report shown in the supplied screenshots does not contain a separate Payments page, although payment data is present in the model. This documentation therefore does not claim a completed Payments page.

Scope note: The project brief also requests drill-through. The supplied screenshots do not provide enough evidence to confirm a completed drill-through workflow, so it is not claimed here as verified.

8. One-Page Stakeholder Summary

Purpose: Provide leadership with a single view of marketplace performance and identify areas for further operational investigation.

Area	Dashboard evidence
Sales	13.59M total sales and 99K orders are shown in the Executive Summary.
Customer value	Average Order Value is 137.75.
Satisfaction	Average Review Score is 4.09 across approximately 98K reviews.
Delivery	94.16% on-time delivery, approximately 96K delivered orders, 8K late orders and 12.41 average delivery days.
Category concentration	Health & Beauty, Watches & Gifts, Bed, Bath & Table, Sports & Leisure and Computers & Accessories appear among the highest-revenue categories.
Geography	Average delivery time varies substantially across states in the displayed geography analysis.

Recommended management follow-up

- Review high-revenue categories for continued sales monitoring and category-level planning.
- Investigate states with longer average delivery times to understand logistics constraints and delivery-estimate accuracy.
- Use the delivery-delay/review-score analysis as a diagnostic starting point for investigating customer dissatisfaction.
- Avoid interpreting the 2017 YoY figure as a full-year growth rate because the 2016 comparison period is incomplete.

9. Limitations and Interpretation Notes

- The dataset covers historical Olist transactions from 2016–2018 and should not be treated as current marketplace performance.
- The dashboard is descriptive: charts identify patterns and differences but do not by themselves establish causality.
- State-level review and delivery comparisons can be affected by differences in order volume and missing delivery/review records.
- The map visual in the supplied screenshot displays a Microsoft mapping/upgrade warning; the map should be verified in the published environment before external presentation.
- The final report contains six pages in the supplied screenshots; a dedicated Payments page was listed in the original brief but is not present in the final screenshots.

10. Portfolio / Resume Description

Olist Marketplace Performance Dashboard | Power BI

Built an interactive Power BI dashboard using the Brazilian E-Commerce Public Dataset by Olist to analyze revenue, orders, product categories, delivery performance, customer reviews, geography and seller performance. Performed data cleaning and transformation in Power Query, designed a dimensional data model, developed DAX measures for sales, time intelligence, delivery and customer satisfaction, and implemented interactive slicers and report-level navigation. The final dashboard provides KPI monitoring and visual analysis of revenue concentration, geographic delivery variation and the relationship between delivery performance and customer reviews.

Document prepared from the supplied Olist project brief and the final dashboard screenshots provided in the project conversation.